

Hands-on Activity 1.6

Programming Essentials

Course Code: CPE 411

Program: Computer Engineering

Course Title: Interactive Extended Reality

Date Performed: 22/01/2026

Section: 22S2

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Instructor: Engr. Jimlord M. Quejado

1. Discussion

The Programming Essentials tutorial introduces learners to the basic concepts of C# programming such as using event loops, variables, and functions such as Instantiate and Delete.

2. Materials and Equipment

Unity Hub and Unity Editor

Visual Studio Code

ChatGPT (for Expert "More things to try")

Learn Unity Website ([Unity Essentials Pathway - Learn Game Development for Beginners | Unity Learn](#))

Computer with Network Connection

3. Procedure

My project - hsa1.4 - Windows, Mac, Linux - Unity 6.3 LTS (60003.3f1)* <DX11>

File Edit Assets GameObject Component Services Jobs Tutorials Window Help

Unity 6 RB Asset Store AI

Hierarchy

- hsa1.4*
 - Environment
 - Cube
 - Directional Light
 - Camera
 - 01_Bedroom
 - Bed_Full_BR001
 - Nightstand_25in_BR001
 - AlarmClock_BR001
 - RugDoorMat_L_Gray
 - RugRectangleLarge_Purple
 - Dresser_30in_BR001
 - LaundryBasket_Round_BR001
 - Rocking Horse
 - RugRectangleLarge_Green
 - BasketBallNet
 - Dresser_55in_BR001
 - Dresser_55in_BR001 (1)
 - Mouse

Scene Game Inspector

Center Global 1 20 Persp

Project Console

Assets

- Unity Essentials
 - Audio
 - Fonts
 - Materials
 - Interior
 - My Materials
 - Outdoor
 - Skyboxes
 - VFX
 - Models
 - Prefabs
 - Bedroom
 - Characters
 - Collectibles
 - Floor
 - Kitchen
 - Lights
 - Living Room

Assets > _Unity Essentials > Prefabs > Characters

- Car_Racing
- Car_Taxi
- DumpTruck
- Mouse
- RobotVacuum_1
- RobotVacuum_2
- Train
- UFO

Tutorials

Add a movement script

Steps 4 / 11 Need Help?

Add a player to the scene

The next step is to add a player to the scene. This player is the character that the user of your game will control.

In this living room scene, your player can be one of the following:

- A robot vacuum sucking up dirt
- A toy car getting stars
- A wind-up mouse finding cheese
- An alien in a UFO collecting gemstones

It's up to you which one sounds the most fun!

Instructions

1. In the Project window, navigate to **_Unity Essentials > Prefabs > Characters** and browse through the options.
2. Click and drag your chosen character into the scene, near the main door on the left.

Warning: Avoid double-clicking the prefab in the Project window because this will open prefab editing mode. Instead, simply drag and drop it into the scene.

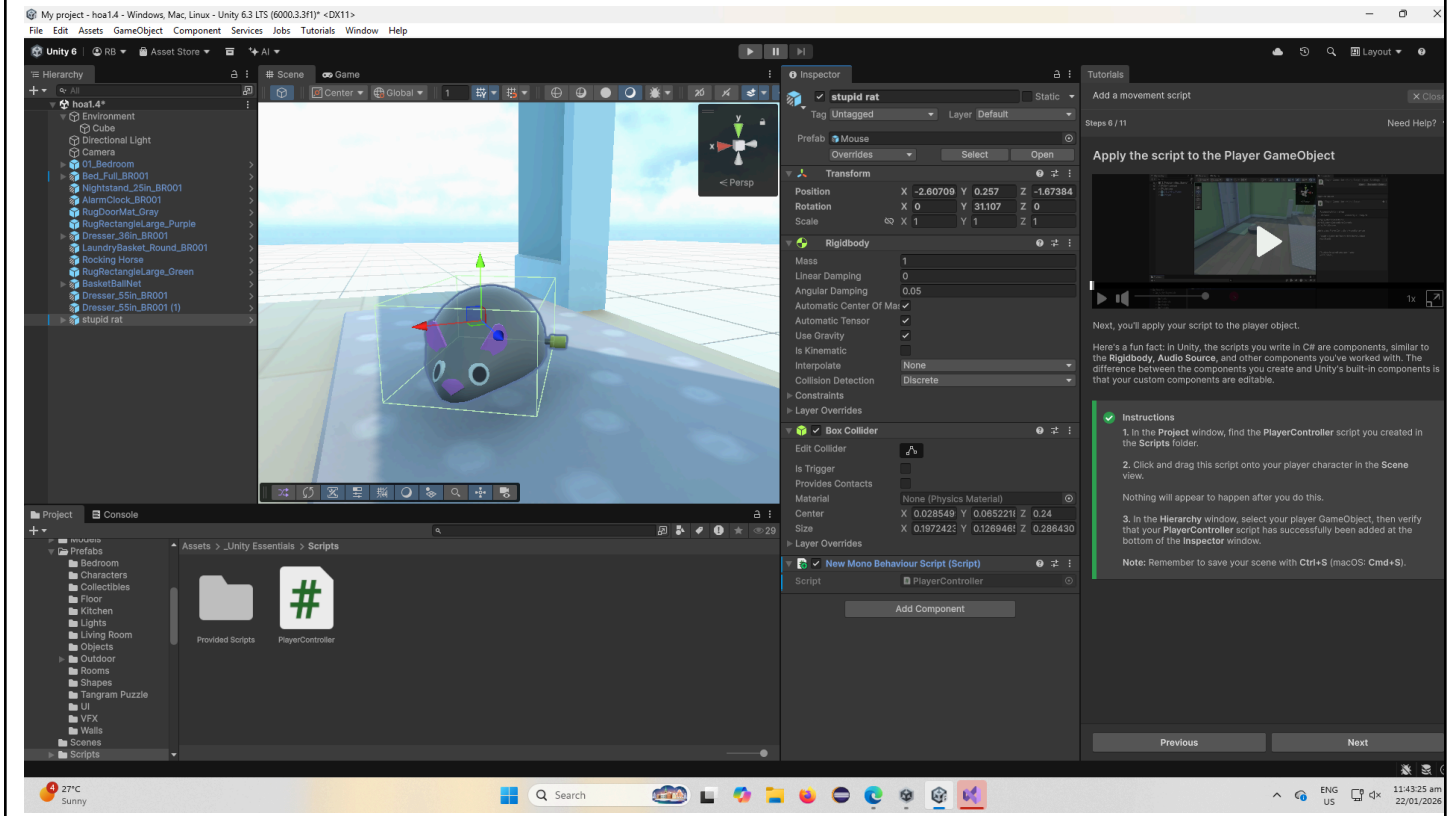
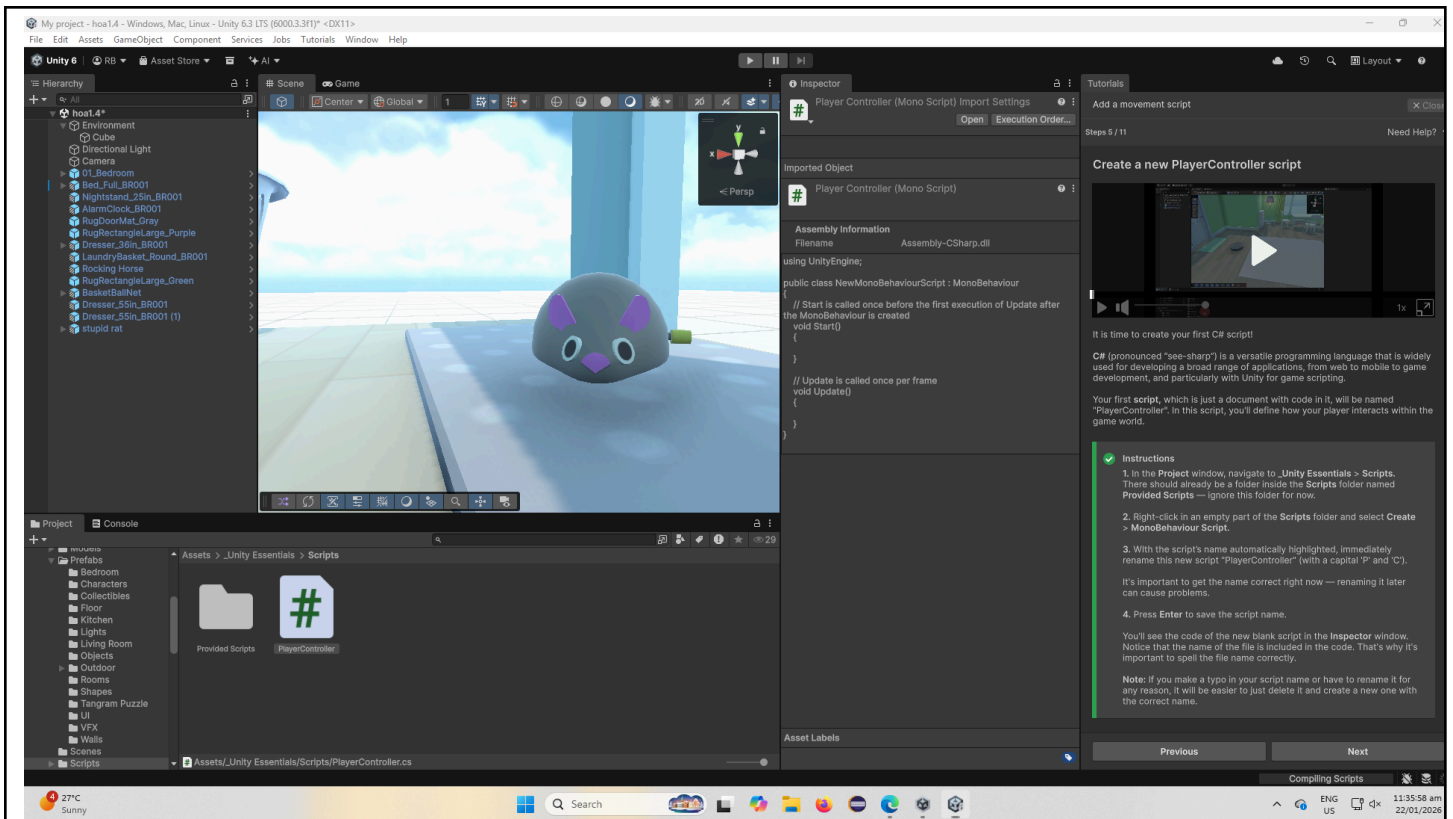
3. If necessary, rotate the player by 90 degrees on the Y-axis so that it's facing the center of the room.
4. In the Hierarchy window, rename your chosen character "Player" for clarity.

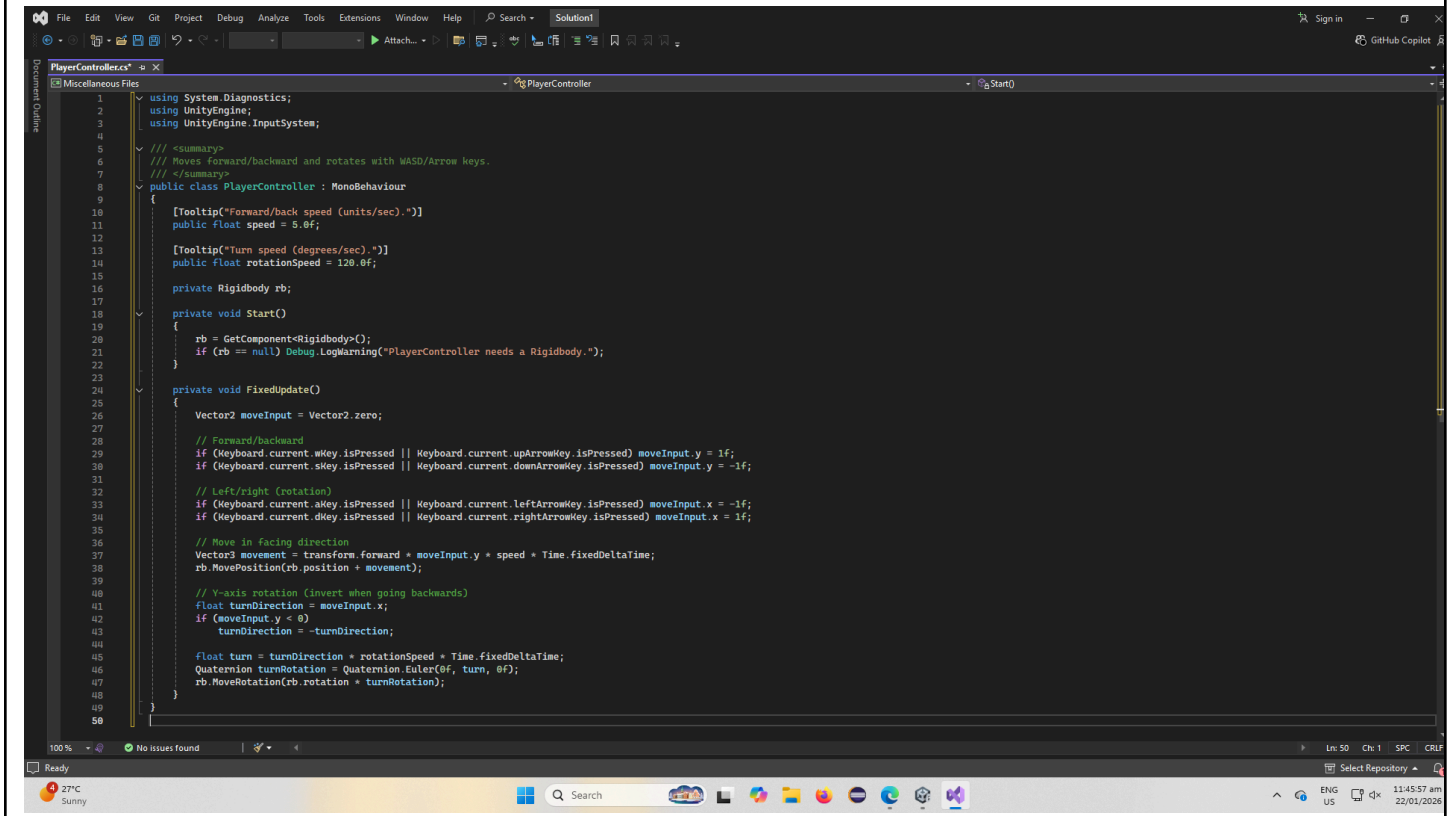
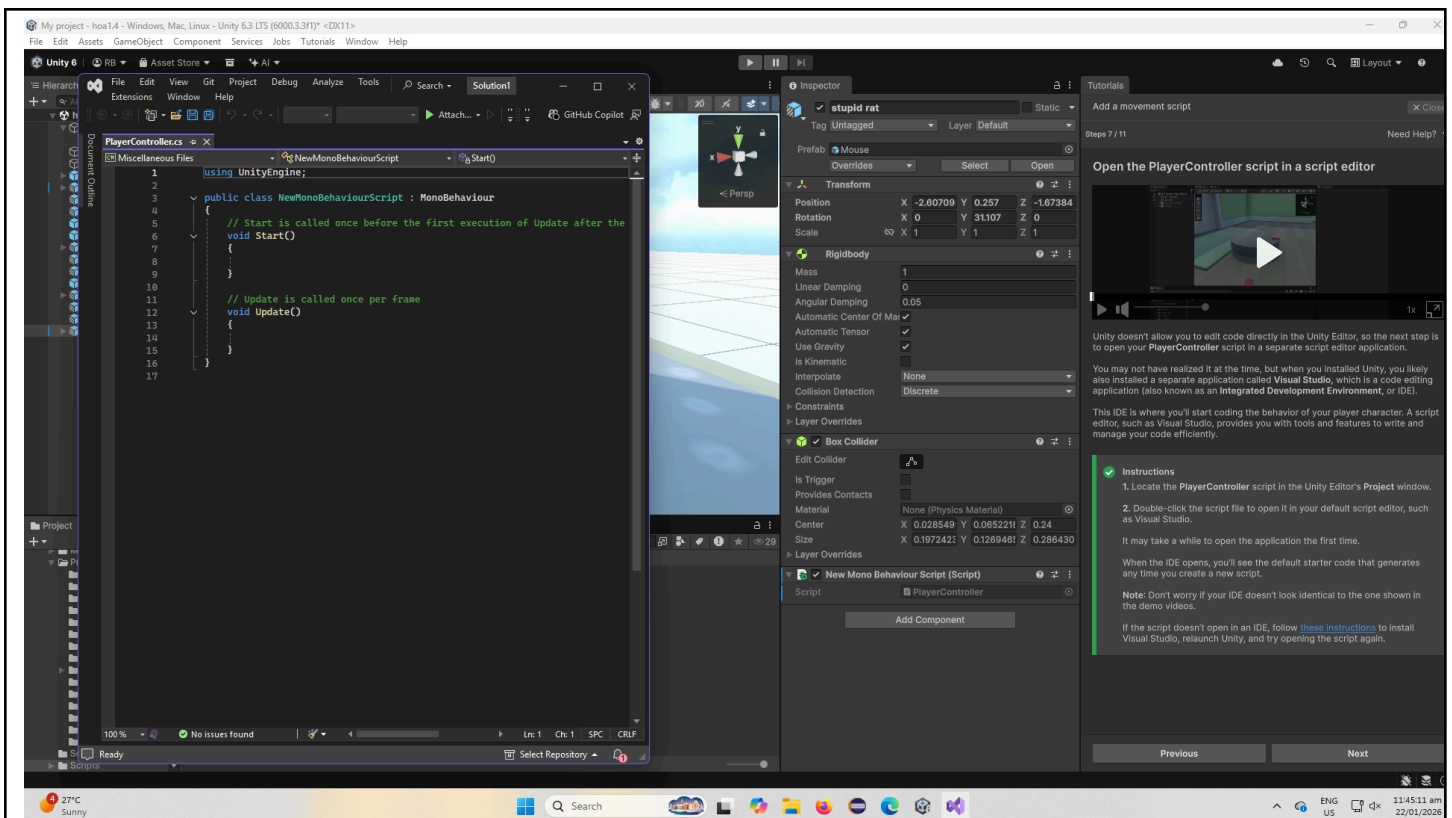
Previous Next

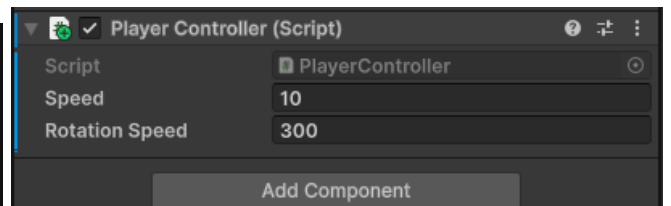
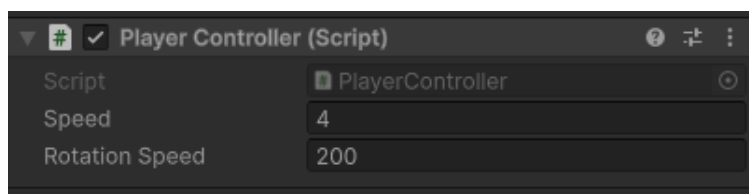
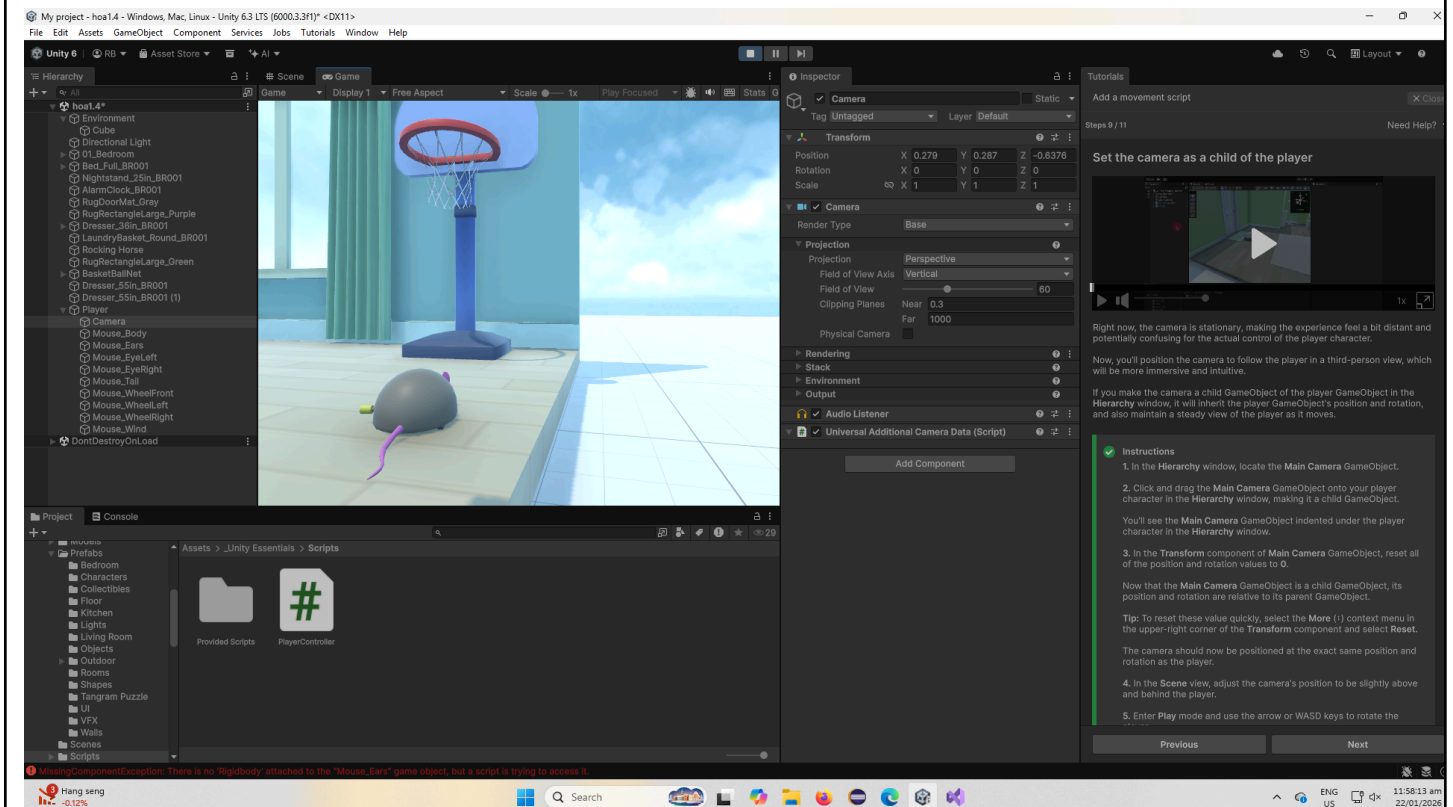
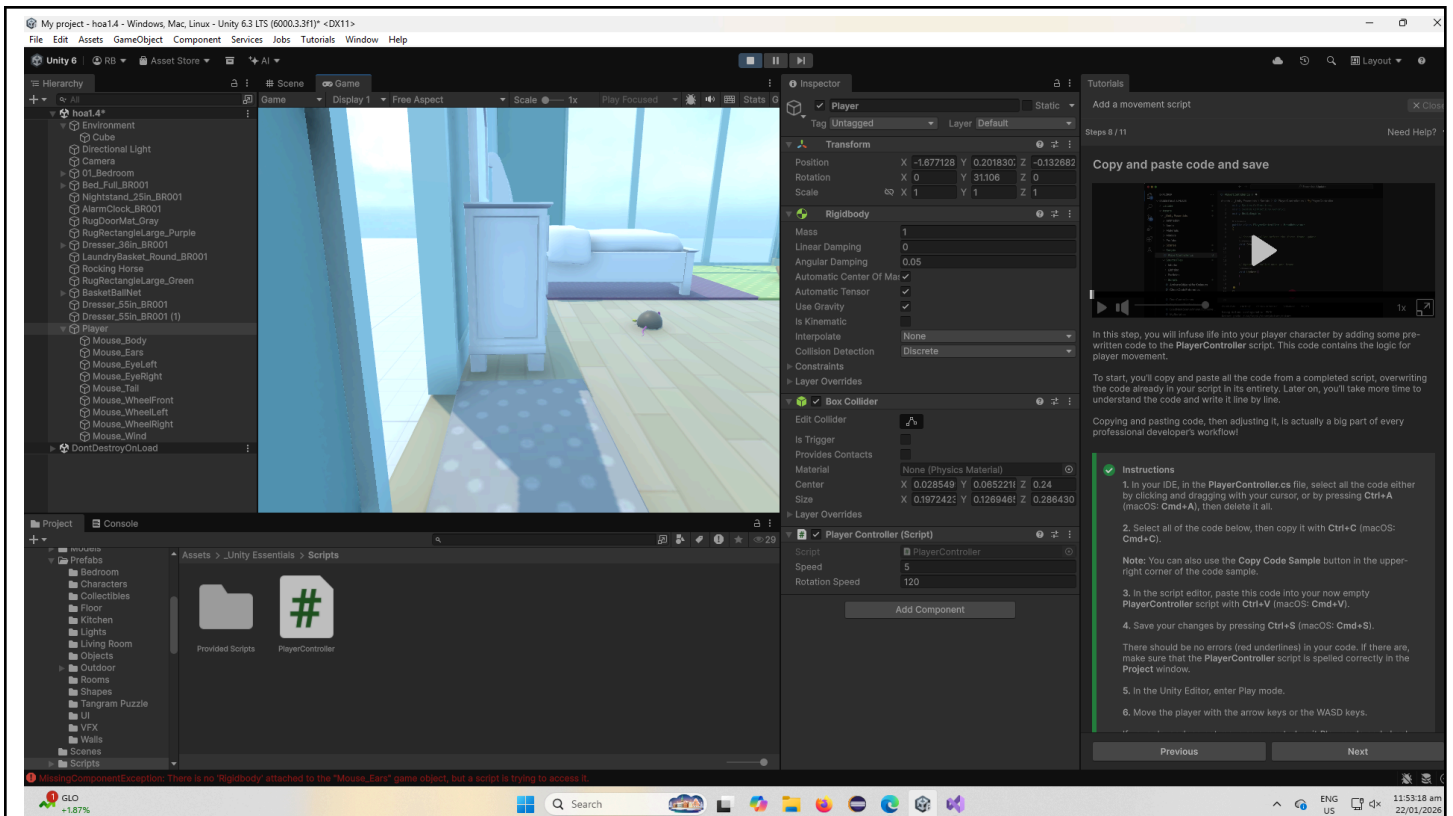
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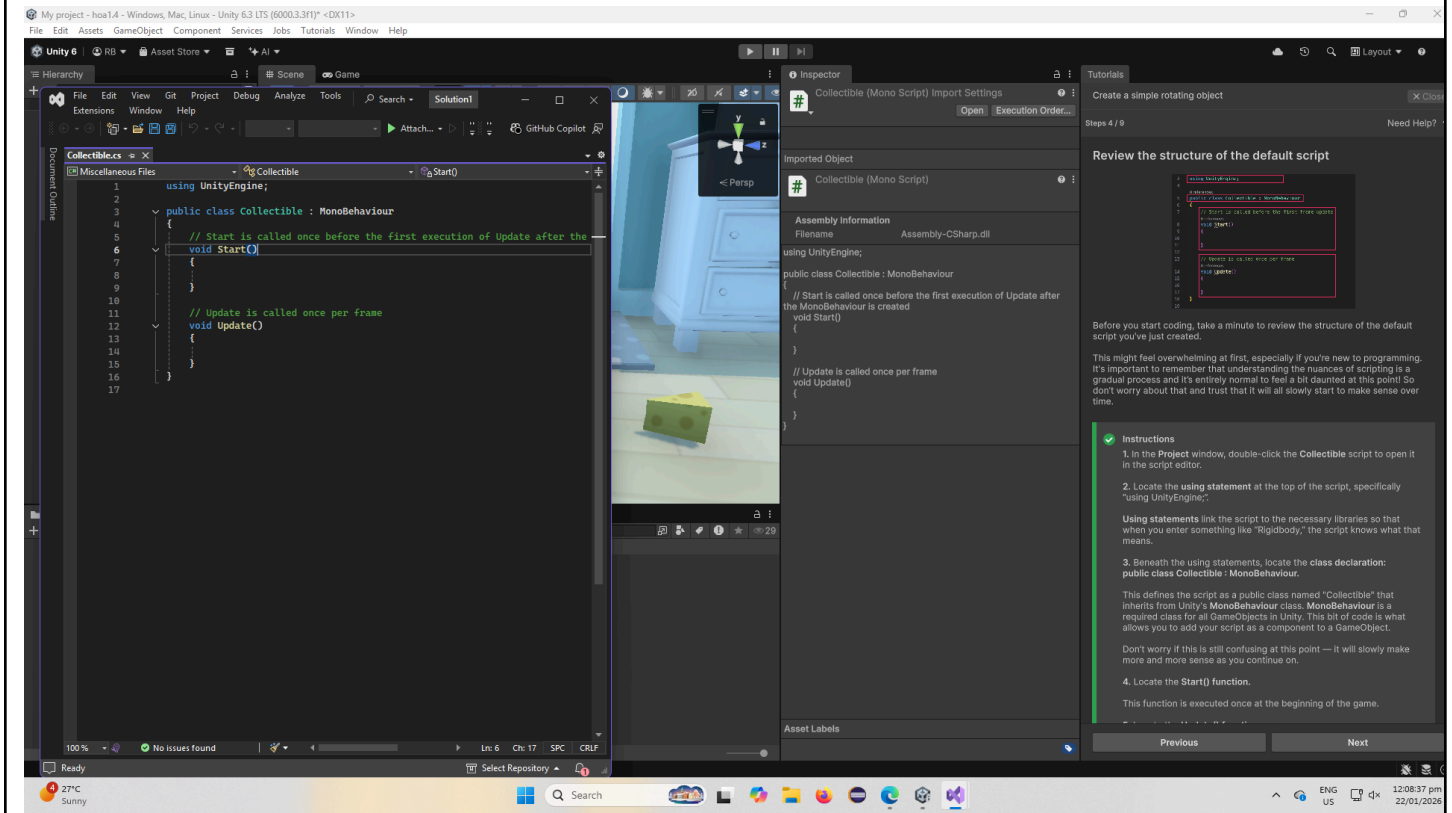
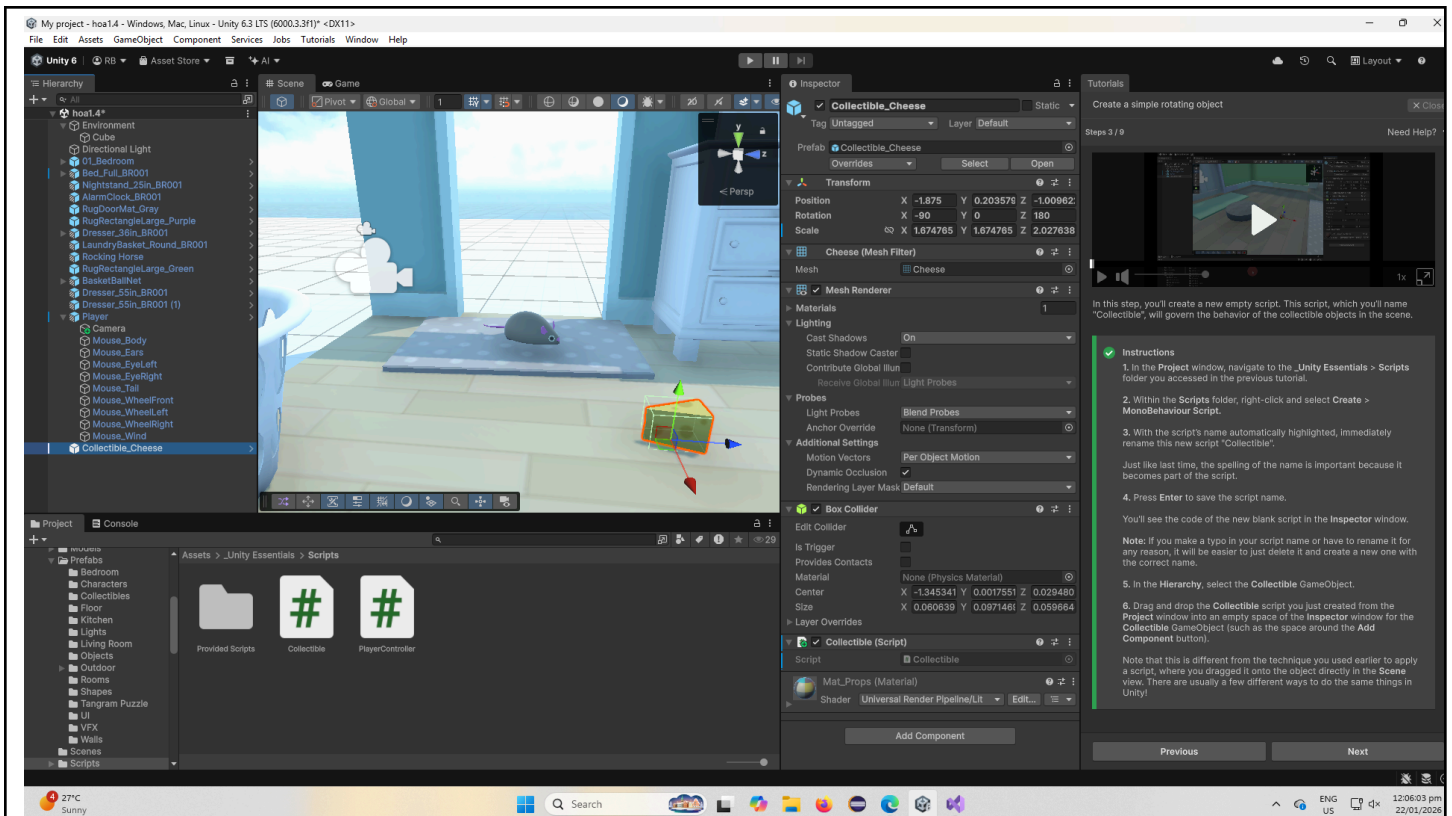
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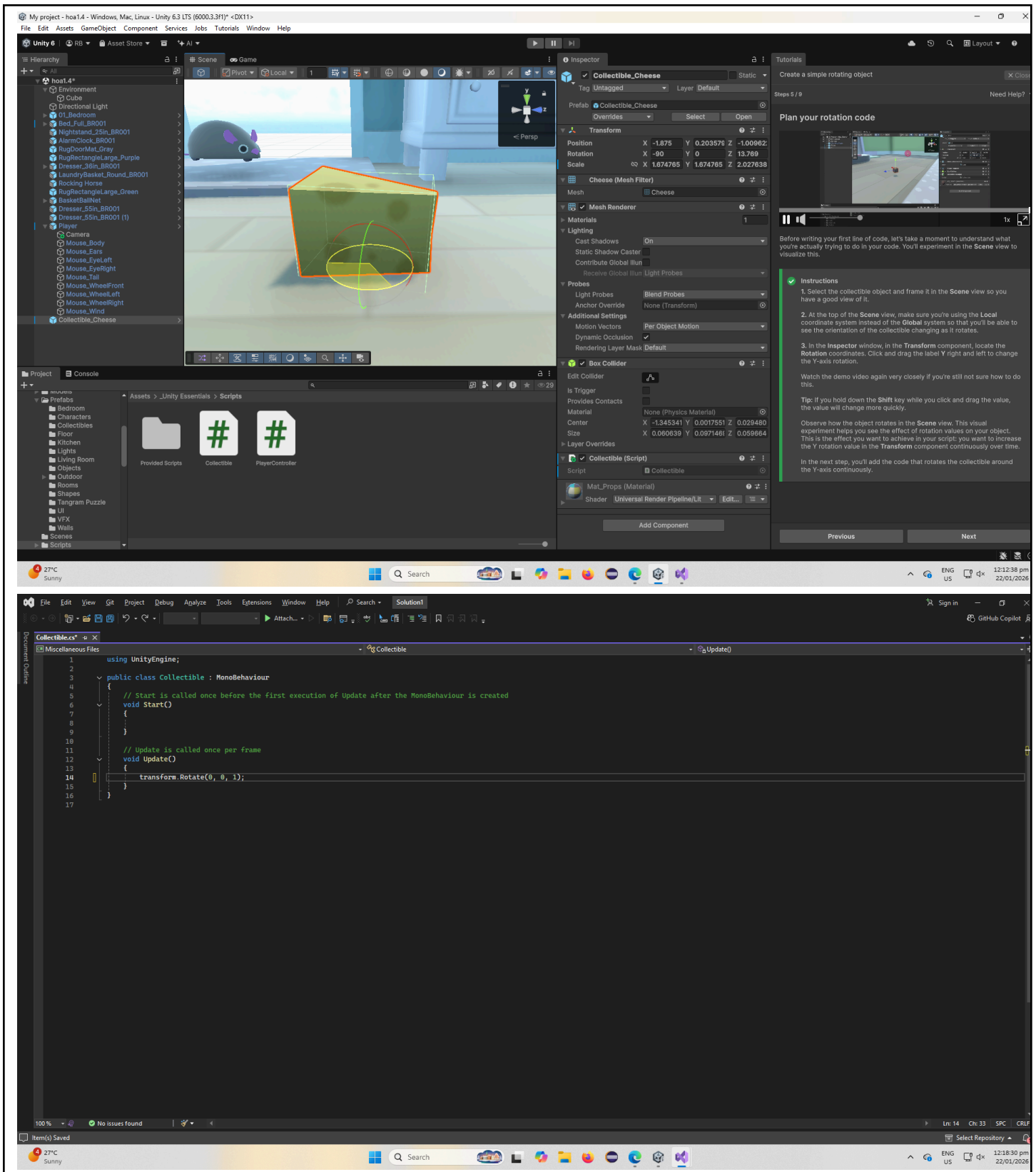
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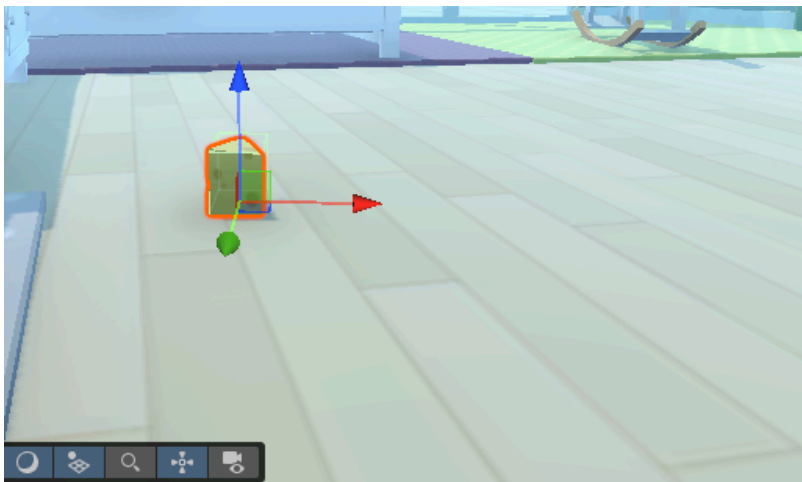








```
File Edit View Git Project Debug Analyze Tools Extensions Window Help Search Solution1
Collectible.cs
1 using UnityEngine;
2
3 public class Collectible : MonoBehaviour
4 {
5     // Start is called once before the first execution of Update after the MonoBehaviour is created
6     void Start()
7     {
8     }
9
10    // Update is called once per frame
11    void Update()
12    {
13        transform.Rotate(0, 0, 0.5f);
14    }
15
16
17
```



Mesh Renderer

- Materials: 1
- Lighting
 - Cast Shadows: On
 - Static Shadow Caster: ☐
 - Contribute Global Illumination: ☐
 - Receive Global Illumination: Light Probes
- Probes
 - Light Probes: Blend Probes
 - Anchor Override: None (Transform)
- Additional Settings
 - Motion Vectors: Per Object Motion
 - Dynamic Occlusion: ☒
 - Rendering Layer Mask: Default

Box Collider

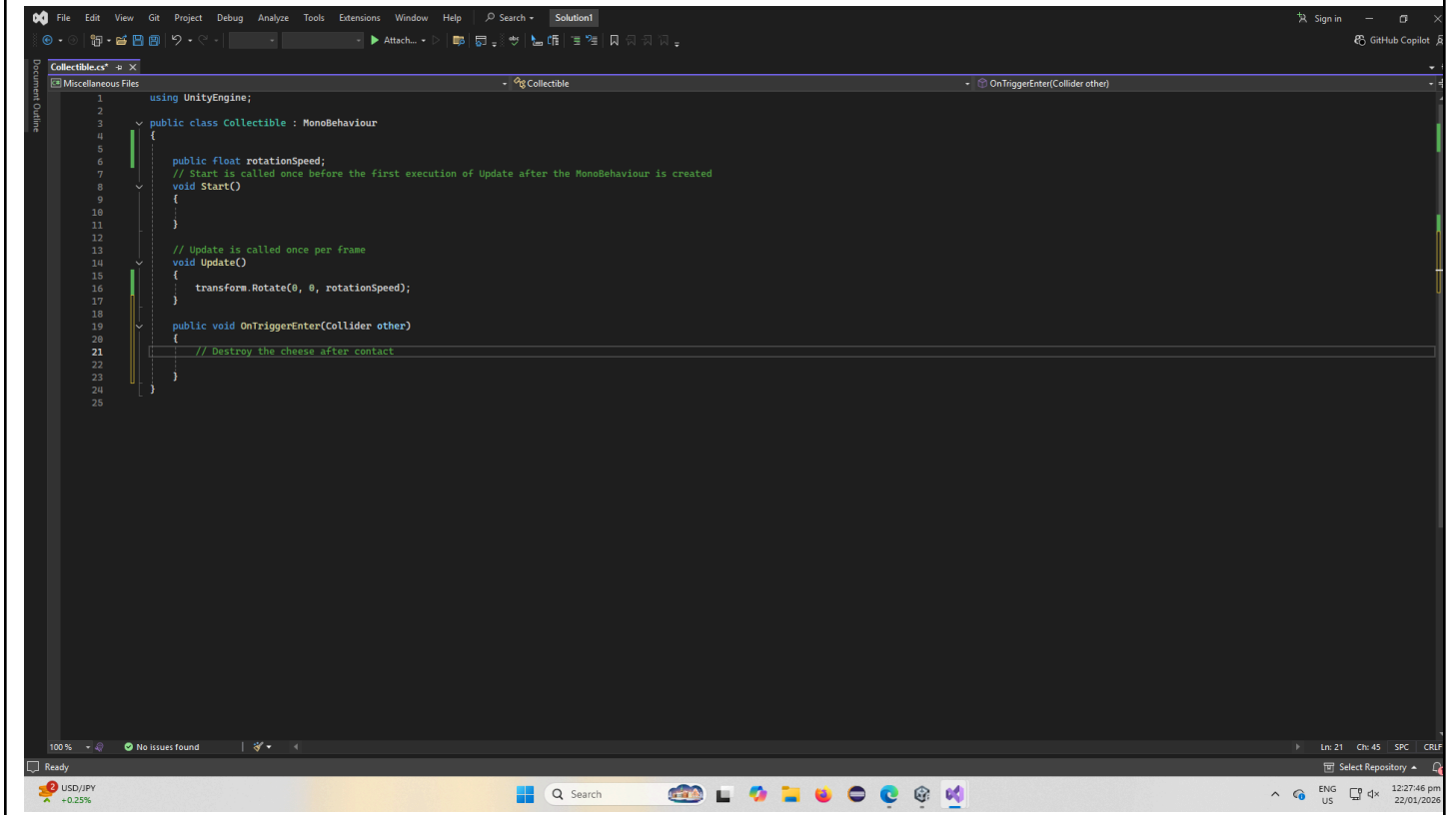
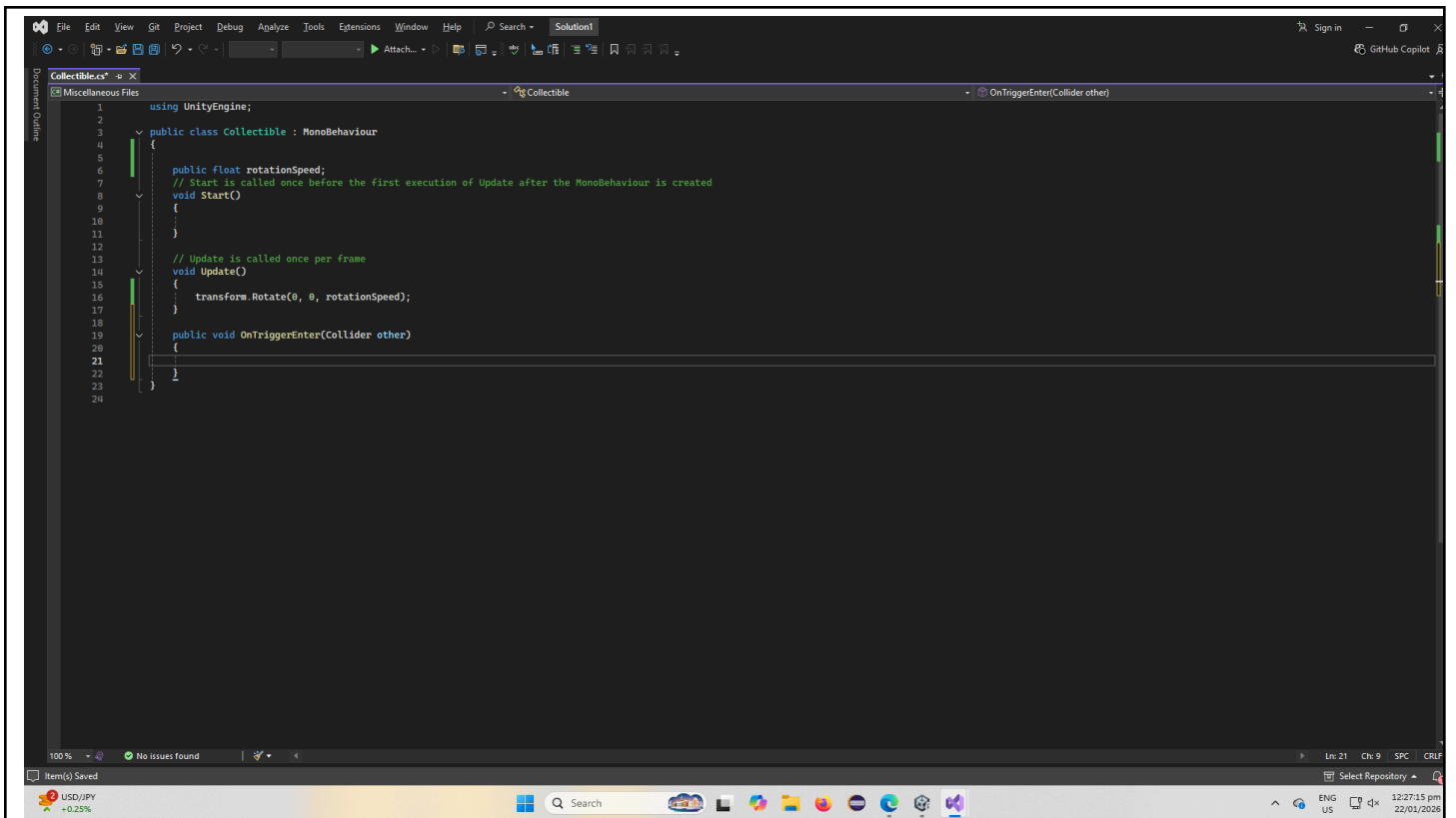
- Edit Collider:
- Is Trigger: ☐
- Provides Contacts: ☐
- Material: None (Physics Material)
- Center: X -1.345341 Y 0.0017551 Z 0.029480
- Size: X 0.060639 Y 0.097146 Z 0.059664
- Layer Overrides

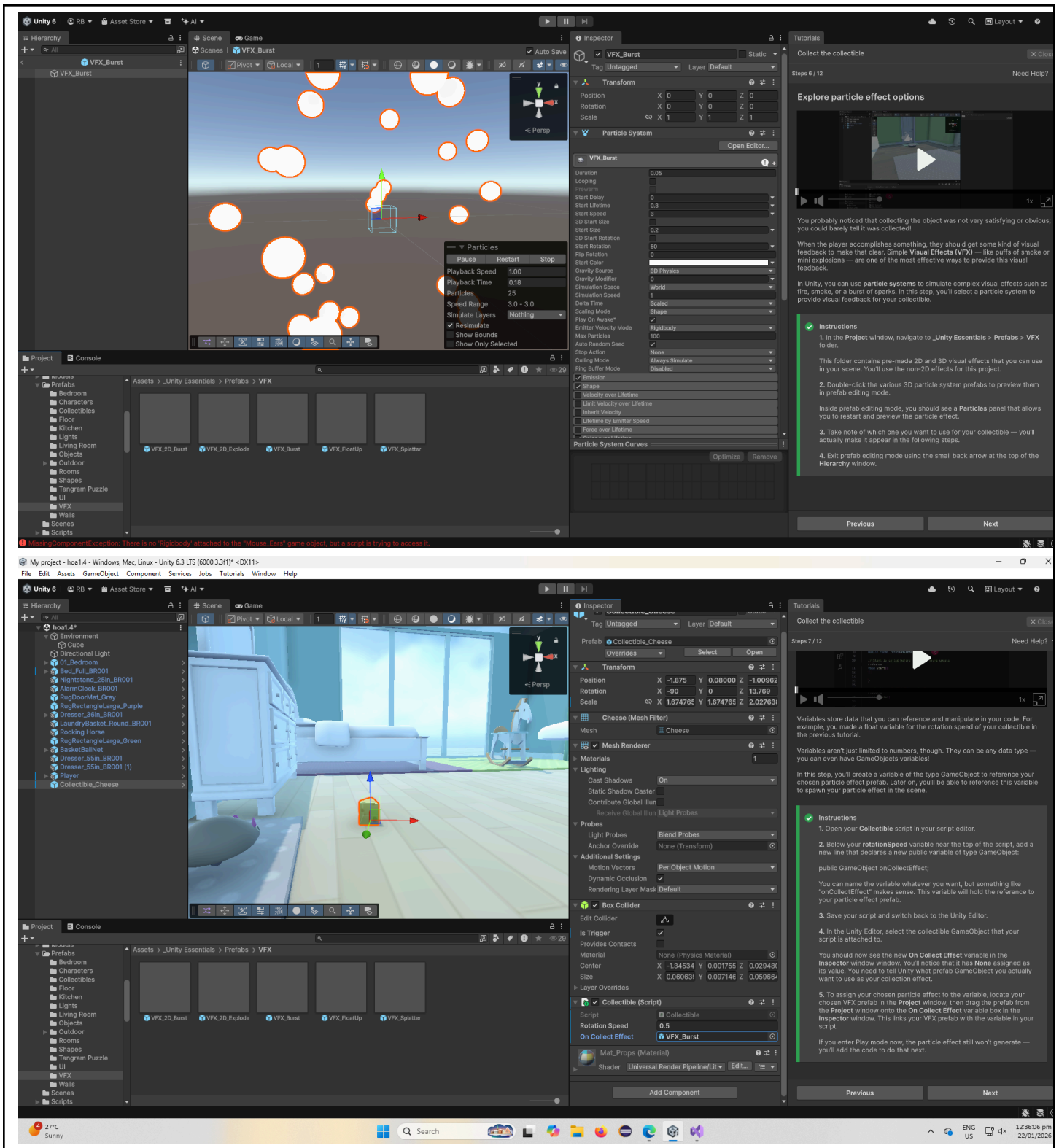
Collectible (Script)

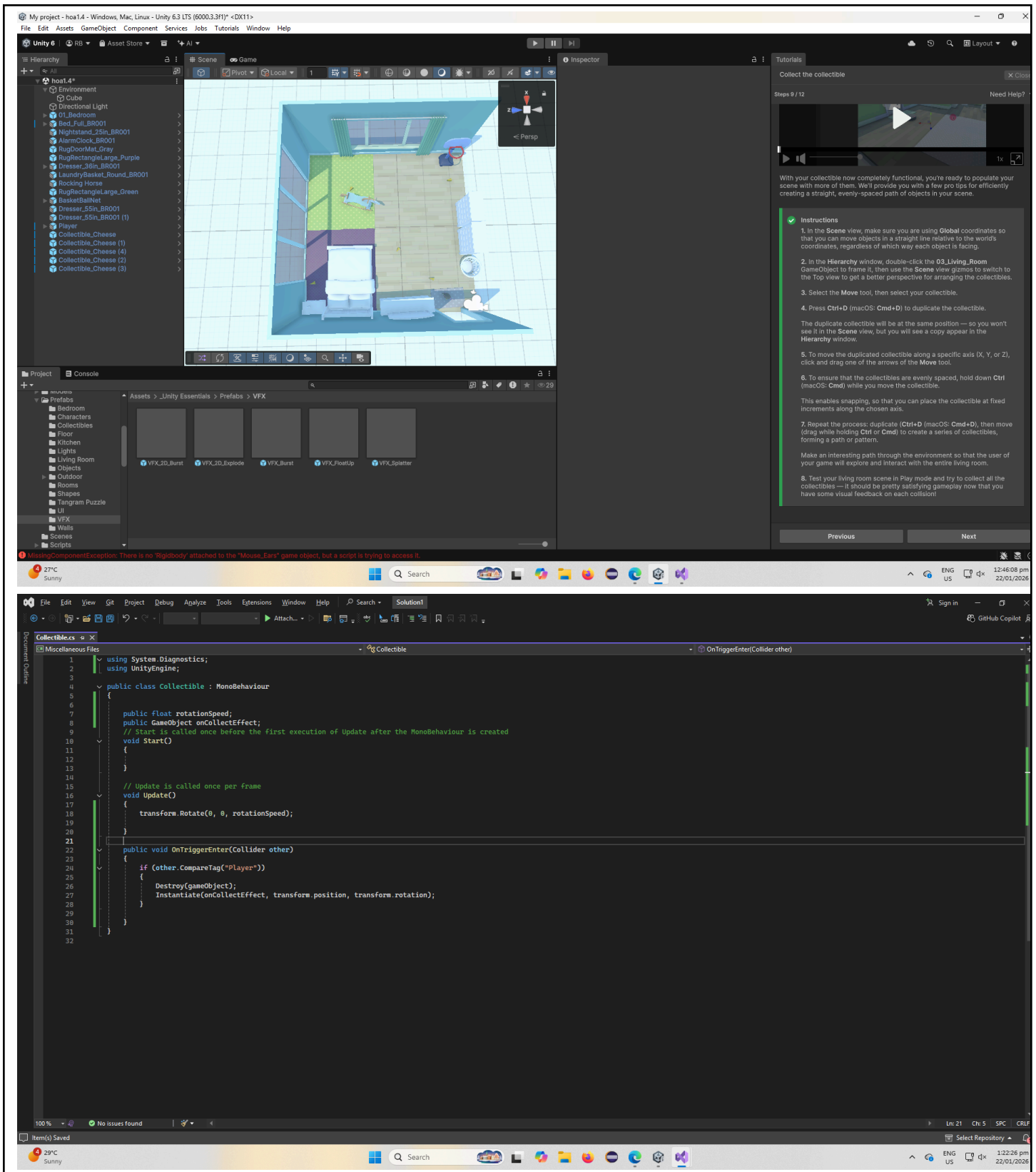
- Script: Collectible
- Rotation Speed: 0.5

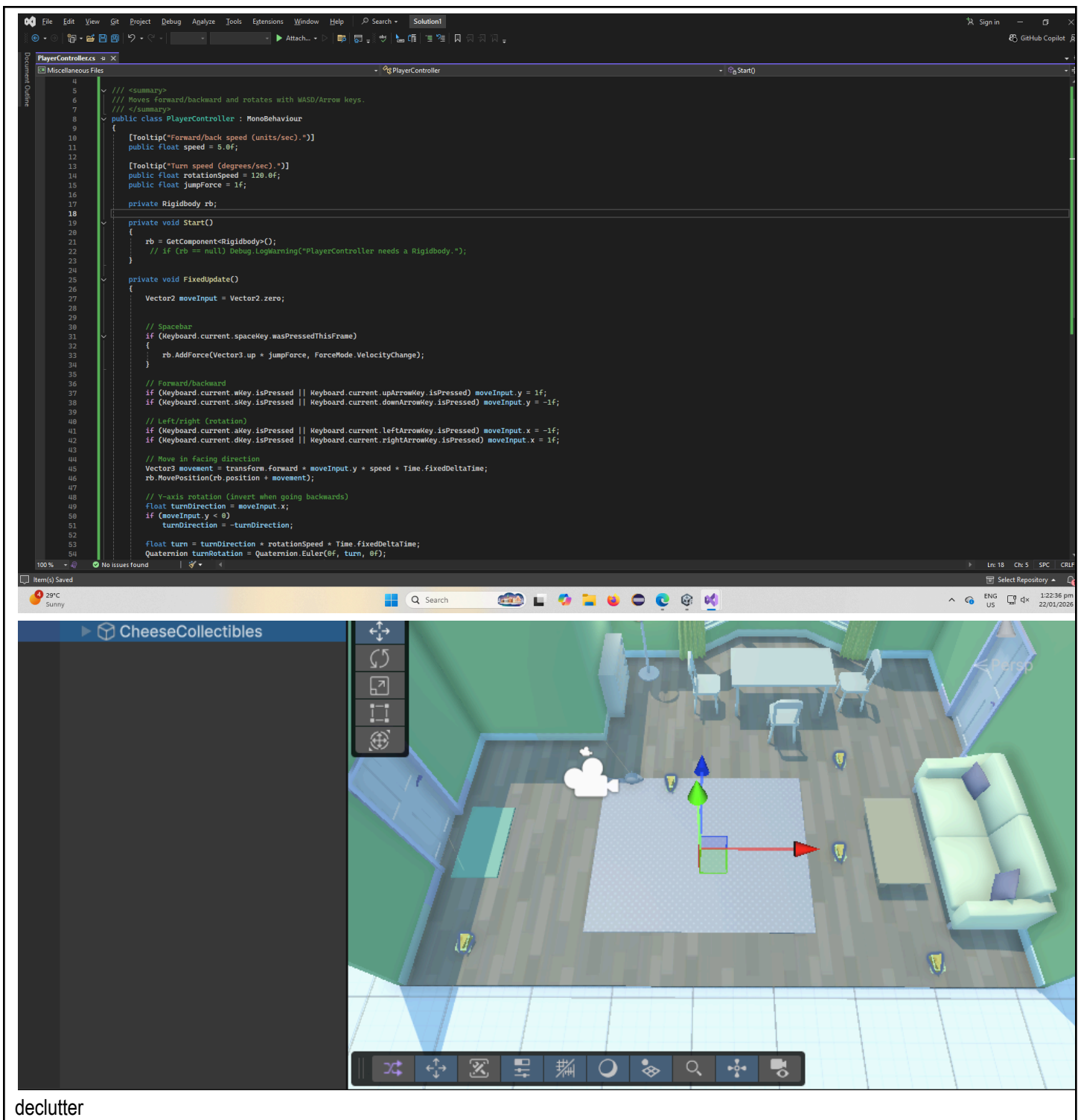
Box Collider

- Edit Collider:
- Is Trigger: ☒
- Provides Contacts: ☐

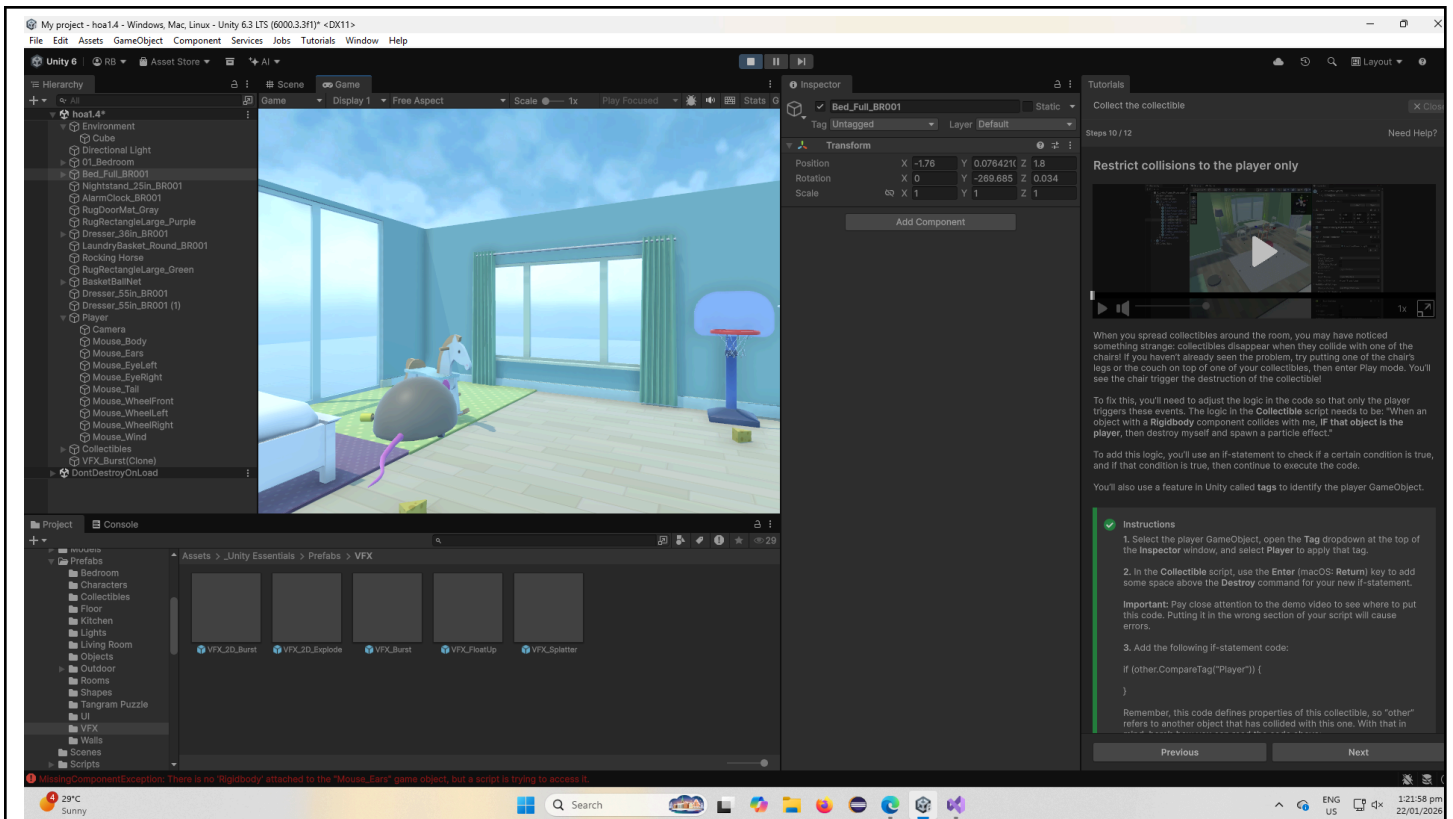




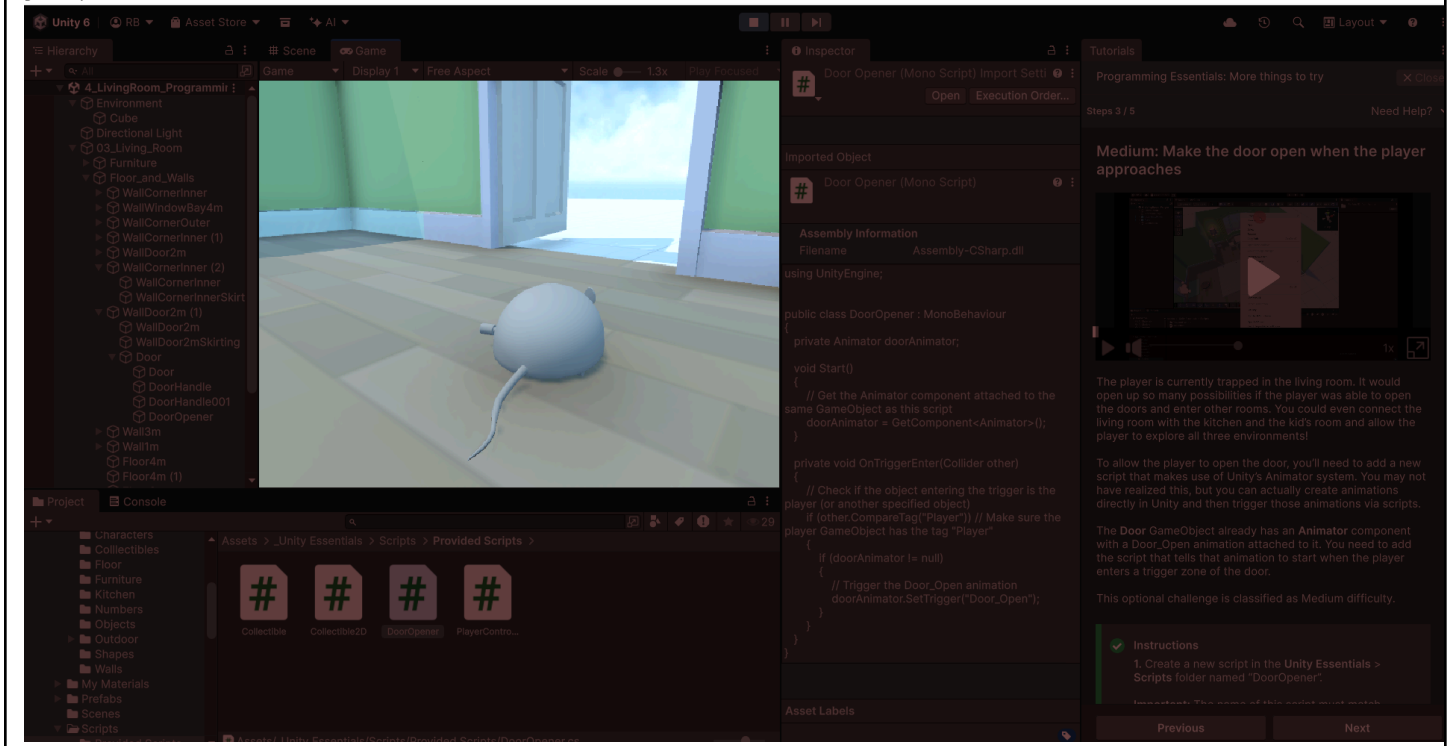


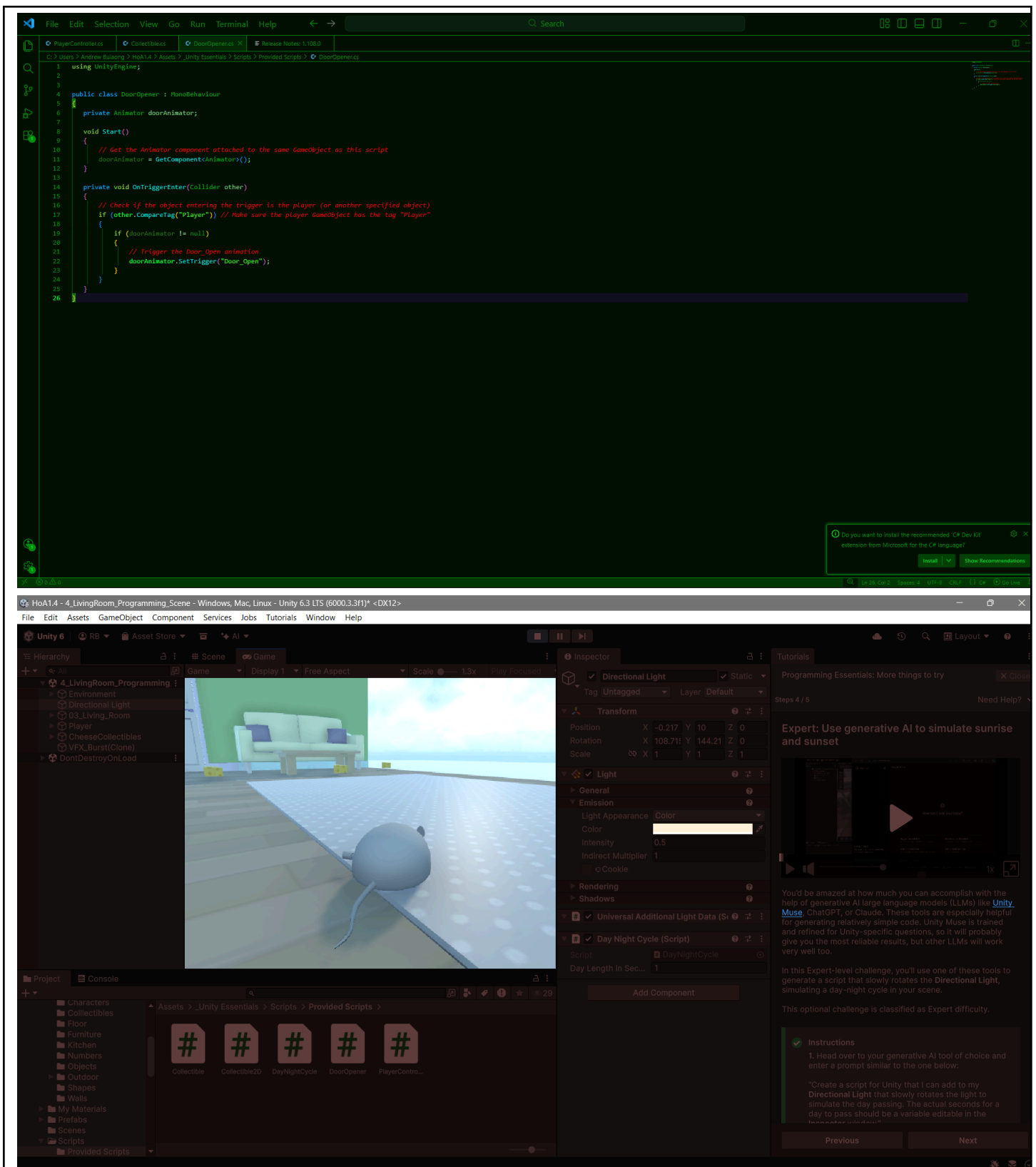


declutter

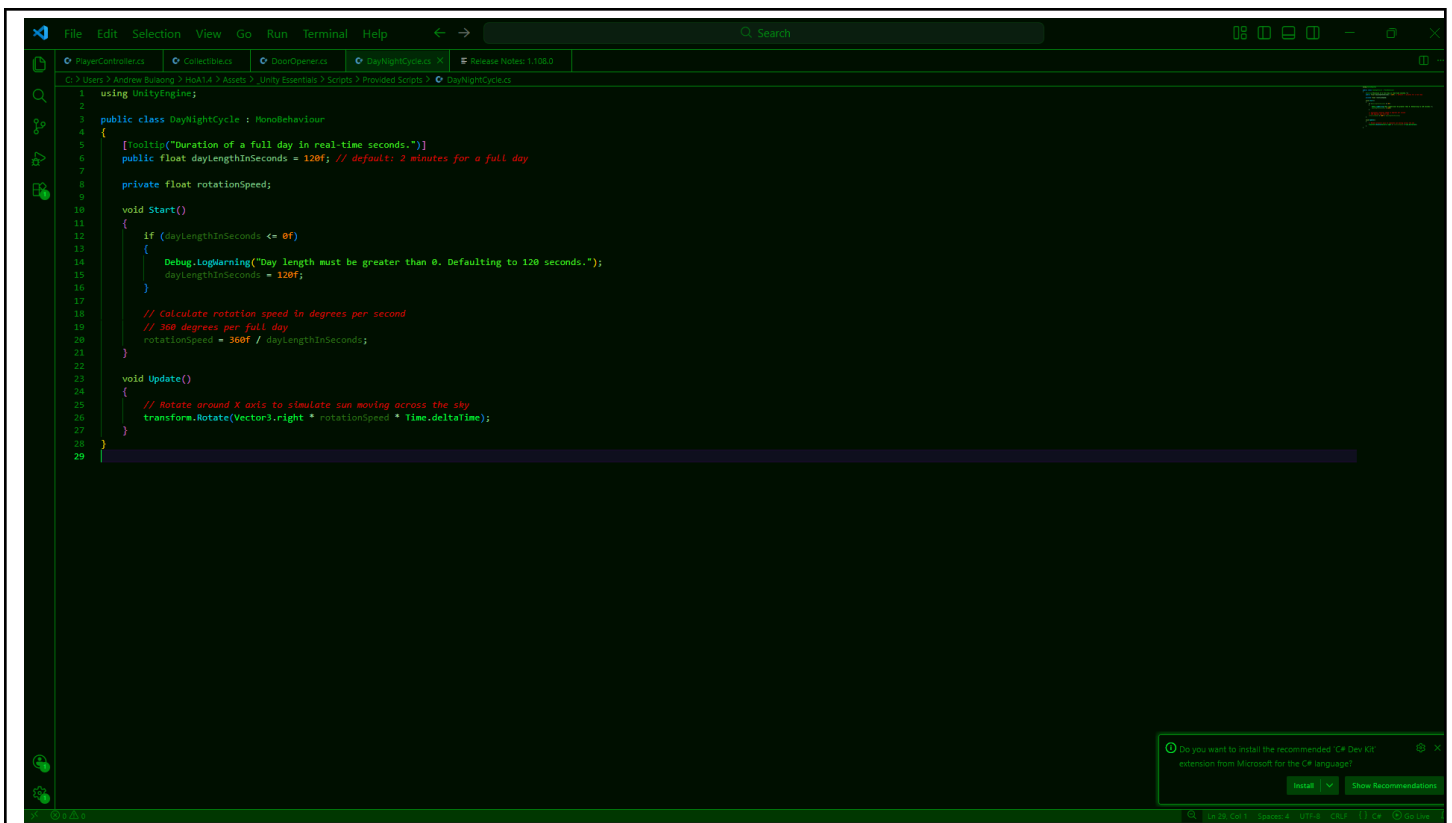


(jump)





day night cycle



```
1 using UnityEngine;
2
3 public class DayNightCycle : MonoBehaviour
4 {
5     [Tooltip("Duration of a full day in real-time seconds.")]
6     public float dayLengthInSeconds = 120f; // default: 2 minutes for a full day
7
8     private float rotationSpeed;
9
10    void Start()
11    {
12        if (dayLengthInSeconds <= 0f)
13        {
14            Debug.LogWarning("Day length must be greater than 0. Defaulting to 120 seconds.");
15            dayLengthInSeconds = 120f;
16        }
17
18        // Calculate rotation speed in degrees per second
19        // 360 degrees per full day
20        rotationSpeed = 360f / dayLengthInSeconds;
21    }
22
23    void Update()
24    {
25        // Rotate around X axis to simulate sun moving across the sky
26        transform.Rotate(Vector3.right * rotationSpeed * Time.deltaTime);
27    }
28 }
29
```

Do you want to install the recommended C# Dev Kit extension from Microsoft for the C# language?

Install Show Recommendations

4. Output

Your results

Congratulations! You passed. If you would like to score higher, you can [retake](#) the quiz. We always save your highest score.

Passed • You got 6 of 6 correct



+60 XP

You gained 60 XP points towards
Unity Essentials

You completed this quiz on January 27, 2026

Next up in Unity Essentials: In-Editor



2D Essentials: In-Editor

+600 XP • 1 Hour 25 Mins

[Continue to next](#)

COMPLETED 

4 | Programming Essentials

Learn the basics of C# to program an experience where the player explores a room to collect objects.



5. Supplementary Activity

No supplementary activity was given.

6. Assessment Rubric

Lab Activity Rubric



Criteria	Ratings						Pts
SO 7 PI 1 Student Outcome 7.1 Acquire and apply new knowledge from outside sources. threshold: 4.8 pts	6 pts Excellent Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently	4 pts Satisfactory Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor Relies on classroom instruction only	1 pts Very Poor No initiative or interest in acquiring new knowledge	6 pts
SO 7 PI 2 Student Outcome 7.2 Learn independently threshold: 4.8 pts	6 pts Excellent Completes an assigned task independently and practices continuous improvement	5 pts Good Completes an assigned task without supervision or guidance	4 pts Satisfactory Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory Requires detailed or step-by-step instructions to complete a task	2 pts Poor Shows little interest to complete a task independently	1 pts Very Poor No interest to complete a task independently	6 pts
SO 7 PI 3 Student Outcome 7.3 Critical thinking in the broadest context of technological change threshold: 4.8 pts	6 pts Excellent Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory Apply the gathered information to formulate the problem	2 pts Poor Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor Gather information from a variety of sources	6 pts
SO 7 PI 4 Student Outcome 7.4 Creativity and adaptability to new and emerging technologies threshold: 4.8 pts	6 pts Excellent Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory Shows some creative ways to solve the problem	2 pts Poor Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor Ideas are copied or restated from the sources consulted	6 pts

Total Points: 24